Evaluate yourself -1

1. An element has a face centered cubic unit cell with a length of 352.4 pm along an edge. The density of the element is 8.9 gcm^{-3}. How many atoms are present in 100 g of an element?
2. Determine the density of CsCl which crystallizes in a bcc type structure with an edge length 412.1 pm.
3. A face centered cubic solid of an element (atomic mass 60) has a cube edge of $4\,\overset{0}{A}$. Calculate its density.

6.6 Packing in crystals:

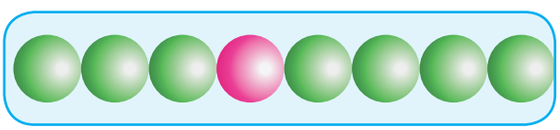
Let us consider the packing of fruits for display in fruit stalls. They are in a closest packed arrangement as shown in the following fig. we can extend this analogy to visualize the packing of constituents (atoms / ions / molecules) in crystals, by treating them as hard spheres. To maximize the attractive forces between the constituents, they generally tend to pack together as close as possible to each other. In this portion we discuss how to pack identical spheres to create cubic and hexagonal unit cell. Before moving on to these three dimensional arrangements, let us first consider the two dimensional arrangement of spheres for better understanding.

6.6.1 Linear arrangement of spheres in one direction:



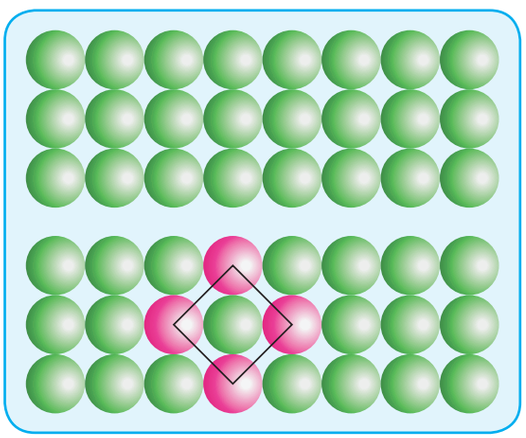
In a specific direction, there is only one possibility to arrange the spheres in one direction as shown in the fig. in this arrangement each sphere is in contact with two neighbouring spheres on either side.

6.6.2 Two dimensional close packing:



Two dimensional planar packing can be done in the following two different ways.

(i) AAA… type:



Linear arrangement of spheres in one direction is repeated in two dimension i.e., more number of rows can be generated identical to the one dimensional arrangement such that all spheres of different rows align vertically as well as horizontally as shown in the fig. If we denote the first row as A type arrangement, then the above mentioned packing is called AAA type, because all rows are identical as the first one. In this arrangement each sphere is in contact with four of its neighbours.

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